

Customer Segmentation Analysis using K-Means

Clustering — Project Report

Oasis Infobyte · Data Analytics Internship

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Project: Customer Segmentation Analysis

1 Introduction

Customer segmentation is the process of dividing customers into groups based on similar characteristics such as demographics, purchasing behaviour, and preferences. Businesses use customer segmentation to better understand their customers and develop personalised marketing strategies.

In this project, **K-Means Clustering** — an unsupervised machine-learning algorithm — was used to identify distinct customer segments from a structured customer dataset.

2 Objectives

- Analyse and understand customer information.
- Clean and preprocess the dataset for machine learning.
- Perform in-depth exploratory data analysis (EDA).
- Identify hidden customer groups using K-Means Clustering.
- Derive actionable business insights from the identified segments.

3 Dataset Description

The dataset contains customer demographic and behavioural information across **14 features**:

Feature	Description
Customer ID	Unique identifier for each customer
Age	Customer age
Gender	Male / Female
Marital Status	Single / Married / Divorced
Education Level	Highest level of education attained
Occupation	Customer's current occupation
Income Level	Annual income bracket
Policy Type	Type of insurance policy
Coverage Amount	Total coverage value
Premium Amount	Monthly/annual premium paid
Customer Preferences	Stated product/service preferences
Preferred Communication	Email / Phone / SMS
Preferred Contact Time	Morning / Afternoon / Evening
Preferred Language	English / Hindi / etc.

4 Tools & Technologies

Python 3.x	Core programming language
Pandas	Data manipulation and analysis
NumPy	Numerical computing
Matplotlib	Data visualisation
Scikit-learn	K-Means Clustering & preprocessing
VS Code	Development environment

5 Project Workflow

1	Dataset Collection	Source and load the customer dataset.
2	Data Exploration	Understand structure, dimensions, and feature types.
3	Data Cleaning	Remove duplicates, handle nulls, standardise columns.
4	Exploratory Data Analysis	Create visualisations to reveal distributions and patterns.
5	Label Encoding	Convert categorical features to numeric values.
6	Feature Scaling	Standardise features using StandardScaler.
7	Elbow Method	Determine optimal K by plotting WCSS vs K.
8	K-Means Clustering	Train the model and assign cluster labels.
9	Cluster Analysis	Calculate group statistics to profile each cluster.
10	Business Insights	Translate cluster profiles into marketing recommendations.

6 Data Exploration

The dataset was explored using standard Pandas operations to understand its structure and quality before any preprocessing.

- Displayed first and last five records (`df.head()` / `df.tail()`)
- Checked dataset dimensions (`df.shape`)
- Examined column names and data types (`df.dtypes`)
- Generated descriptive statistics (`df.describe()`)
- Identified missing values (`df.isnull().sum()`)

- Checked for duplicate records
- Counted unique values per column (*df.nunique()*)

Observation: The dataset was successfully loaded. Its structure and feature types were fully understood, enabling informed preprocessing decisions.

7 Data Cleaning

- Removed duplicate records to prevent biased clustering.
- Handled missing values via imputation or row removal as appropriate.
- Standardised column names (lowercase, underscores).
- Removed unnecessary whitespace from string columns.
- Verified and corrected data type mismatches.

Observation: The dataset was cleaned and prepared. All quality checks passed, confirming readiness for machine learning.

8 Exploratory Data Analysis (EDA)

Multiple visualisations were produced to uncover patterns in customer demographics and financial behaviour:

Visualisation	Type	Key Insight
Age Distribution	Histogram	Most customers aged 30–50
Gender Distribution	Bar Chart	Balanced gender split
Education Levels	Bar Chart	Graduate customers dominate
Policy Type Split	Pie Chart	Term policies most common
Premium Distribution	Histogram	Right-skewed; high earners
Avg Premium by Age	Line Chart	Premium peaks at 45–55

Observation: Visualisations revealed clear demographic and financial patterns, providing a strong foundation for the clustering phase.

9 Feature Engineering

K-Means relies on distance calculations and therefore requires numeric input. **Label Encoding** was applied via Scikit-learn's *LabelEncoder* to convert all categorical columns into integer representations.

Observation: All categorical variables were successfully transformed. No information was lost during encoding.

10 Feature Scaling

StandardScaler was applied to the entire feature set. Scaling ensures that features with large numerical ranges (e.g., Income, Coverage Amount) do not dominate the Euclidean distance calculations used by K-Means.

Observation: All features were standardised to a mean ≈ 0 and standard deviation ≈ 1 , ensuring equal contribution during clustering.

11 Elbow Method

The **Within-Cluster Sum of Squares (WCSS)** was computed for $K = 1$ to 10. The plot was examined to identify the *elbow point* — the value of K beyond which additional clusters yield diminishing returns.

K	WCSS (relative)	
1		
2		
3		
4		← Elbow
5		
6		
7		
8		
9		
10		

Observation: The elbow point was clearly observed at $K = 4$. Therefore, four clusters were selected for customer segmentation.

12 K-Means Clustering

The K-Means algorithm (`sklearn.cluster.KMeans`) was trained with $K = 4$ on the scaled dataset. Cluster labels were appended to the original dataframe for downstream analysis.

- Algorithm: K-Means (Lloyd's algorithm)
- Initialisation: k-means++ for faster convergence

- Max iterations: 300
- Random state: fixed for reproducibility

Observation: The algorithm converged successfully. Each customer was assigned to one of four distinct clusters based on feature similarity.

13 Cluster Analysis

Group statistics (mean values of numeric features) were computed per cluster to understand the profile of each customer segment:

Cluster	Profile Label	Characteristics
	Young Professionals	Lower income, basic policy, digital-first
	Mid-Career Earners	Moderate income, mixed policy types
	High-Value Clients	High income, comprehensive coverage
	Senior Customers	Older demographic, long-term policies

Observation: The four clusters represent distinct customer profiles based on demographic and behavioural attributes, enabling highly targeted marketing strategies.

14 Business Insights

Cluster 0 — Young Professionals

Focus on affordable entry-level products, mobile-first digital channels, and social media campaigns.

Cluster 1 — Mid-Career Earners

Offer policy upgrades and bundled family plans; engage via email and weekend campaigns.

Cluster 2 — High-Value Clients

Prioritise premium, personalised service; assign relationship managers and offer exclusive benefits.

Cluster 3 — Senior Customers

Highlight long-term security, phone-based support, and loyalty rewards for retention.

Observation: Customer segmentation enables precision marketing, improved retention, and better allocation of sales and marketing budgets.

15 Conclusion

This project successfully implemented an end-to-end customer segmentation pipeline using the K-Means Clustering algorithm. The complete data analytics workflow was executed, from raw data exploration through to actionable business insights:

- Data exploration and cleaning ensured high-quality input data.
- EDA uncovered meaningful patterns in demographics and behaviour.
- Feature engineering and scaling prepared the data for machine learning.
- The Elbow Method identified $K = 4$ as the optimal cluster count.
- Four distinct customer segments were successfully identified and profiled.
- Business insights were derived to guide personalised marketing strategies.

16 Future Scope

- Compare K-Means against Hierarchical Clustering and DBSCAN for robustness.
- Build an interactive dashboard using Power BI or Tableau for stakeholder reporting.
- Deploy the segmentation model as a REST API or web application.
- Integrate Customer Lifetime Value (CLV) prediction alongside segmentation.
- Implement real-time customer segmentation using streaming data pipelines.

I would like to thank **Oasis Infobyte** for providing this internship opportunity and for the guidance extended throughout the project. This work has greatly enhanced my practical skills in data analytics, machine learning, and professional reporting.